



Charging Ahead

*The next wave
of US power
consumption*

September 18, 2024

GQG
Research



Key Takeaways

- > After a period of stagnation, electricity demand in the US is set to increase significantly due to the expansion of data centers driven by AI technology, widespread EV adoption, and the overall electrification of the economy
- > Power-hungry facilities like data centers are moving away from traditional hubs, creating demand in more geographically diverse areas
- > Each EV that replaces an internal combustion engine increases electricity consumption by 80 kWh; this shift could represent a significant percentage of the projected grid capacity by 2040



The era of flat power demand is over. The rise in electricity needs by data centers, low carbon technologies, and electric vehicle (EV) adoption across the US has further increased near-term electricity demand forecasts.

Electricity demand growth in the US remained flat for the past two decades after growing at a 5 percent rate for the second half of the 20th century. Improvements in energy efficiency and outsourcing of major industrial operations to developing countries kept load growth unchanged since the

early 2000s.¹

Yet major structural changes are occurring that require massive investments in the power sector where electricity demand is expected to grow by at least 2 percent each year for the next two decades.¹ The ramp-up in demand is being driven by the rise in data centers and AI, a fuel shift to clean energy, and the electrification of the economy.

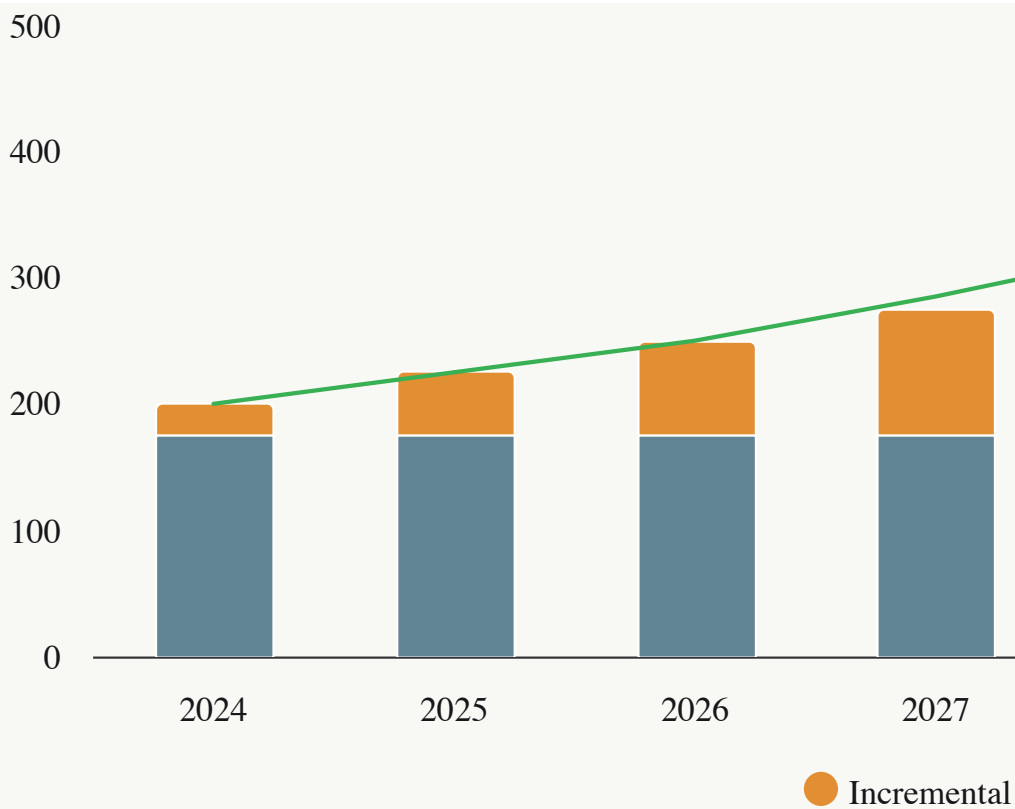
| HYPERSCALE DATA CENTERS

In our view, investing in the power and utility sectors is a way to play the AI boom beyond semiconductors and software companies.

A new wave of data center power demand—accelerated by generative artificial intelligence (GenAI)—is 4x to 5x that of traditional data centers, which historically have been concentrated in Loudon County, Virginia.²

The most recent surge in electricity demand from hyperscalers is poised to grow at a 10 percent rate through 2030 and could triple from 2.5 percent in 2022 to 7.5 percent of total electricity consumption in the US. That is the equivalent of the electricity used by about 40 million US homes.³

US Electricity Demand from Datacenters (TWh)



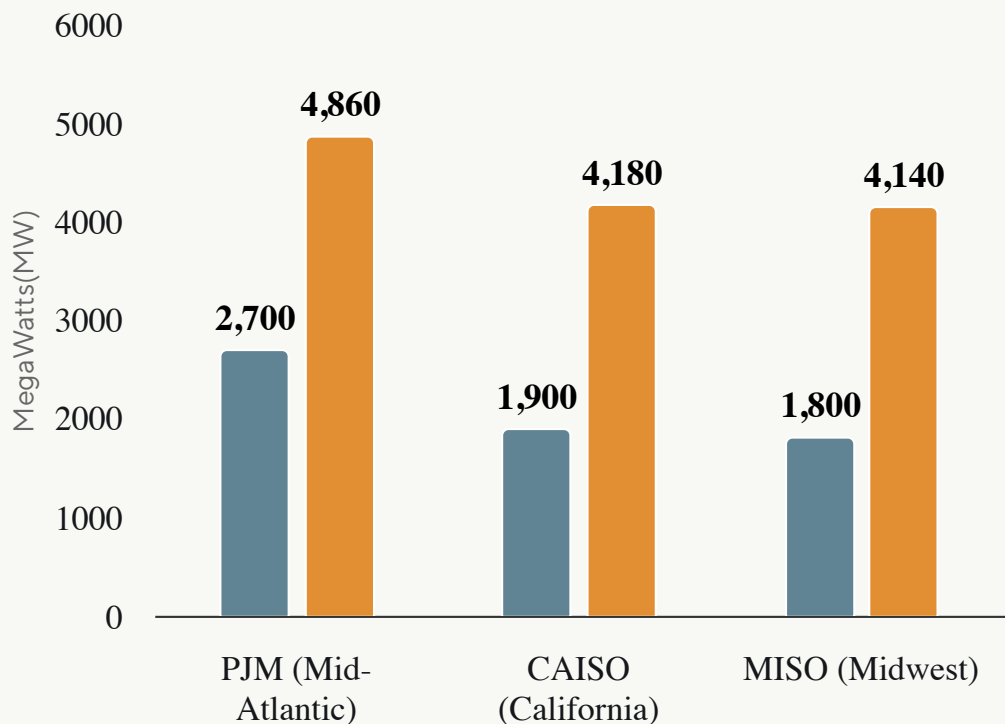
Sources: S&P Global Commodity Insights; 451 Research (historical). Data compiled January 29, 2024. Utility forecast includes data center forecasts from Pennsylvania-New Jersey-Maryland (PJM) (based on utility forecasts),

“As AI produces a lot of load growth, utilities aren’t positioned to put a lot of power facilities in place quickly,” said a former infrastructure planning executive at US’ PJM Interconnection. “Utilities and regulators are not preparing well and are already way behind. Of course, that depends on growth of different load segments in any given part of the country.”

The placement of data centers is becoming more geographically diverse, requiring expanding the grid to remote areas and away from the traditional I-66 corridor in northern Virginia. It used to be that data centers needed to be close to the backbone of the internet, near Washington DC, because the internet originated from the federal government. But now data centers are being constructed in California, Ohio, Illinois, Georgia, and Florida, among other

regions.

>60% of Total US Data Center Capacity Expected in PJM, MISO, CAISO and Southeast by 2027



Sources: GQG Partners. Digital Infra Real Estate. Omidia, Arizton, BCG

FUEL SHIFTING TO CLEAN ENERGY

Utilities are increasingly shifting their generation mix from coal to renewables and nuclear energy. Renewable energy generates over 20 percent of all US electricity and is expected to more than double in 15 years. To make the switch to renewables from fossil fuels, the US will need 3x the renewable generation capacity due to the intermittency of solar and wind sources.⁴

“Texas has about 40,000 megawatts (MWs) of renewables installed, almost half of what its peak demand was last summer of 85,000 MWs. In Texas, where renewables represent almost half of the generation fleet, the

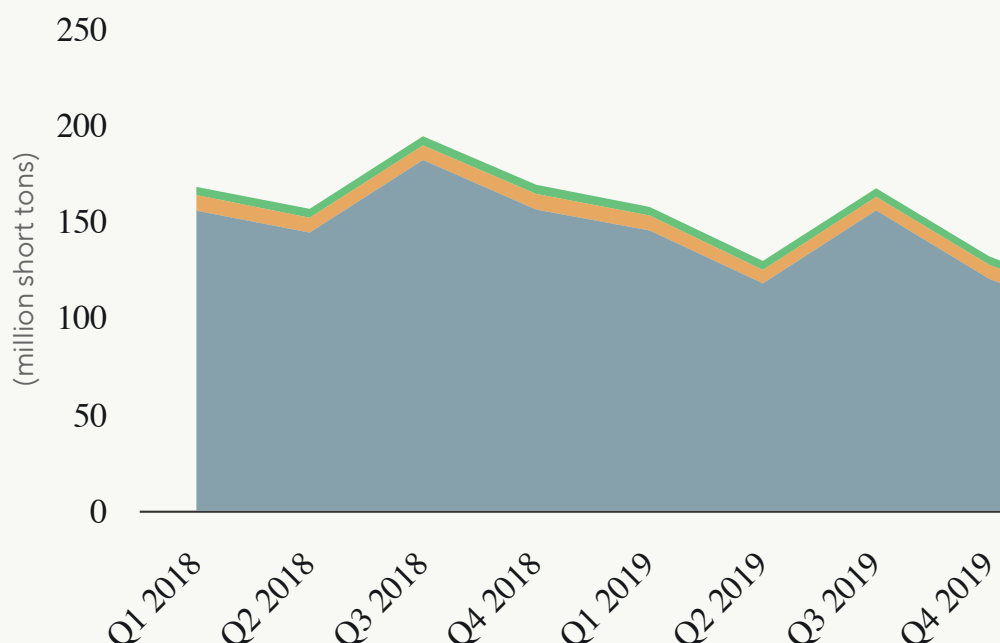
intermittency issues are large. During the snowstorm of Christmas 2022, windmills couldn't turn in those conditions, and solar panels were covered with snow, so those transmission poles and those solar panels sat idle," said a former executive at the Public Utility Commission of Texas.

Thus, demand for fossil fuels is expected to be more volatile—lower on average but potentially much higher on peak—on days when intermittent renewables are at low generation levels.

Coal demand from the US power sector has declined substantially in absolute terms in recent years, yet the sector is expected to consume 73 percent of US coal this year and 70 percent in 2025.⁵

Total Annual Consumption of U.S. Coal

Jan 2018 – Jan 2024



Source: U.S. Energy Information Administration, Quarterly Coal Report,

Nuclear power plants have generated about 20 percent of

US electricity since 1990.⁶ Prior to 2023, only one new nuclear reactor had come online since 1996. That changed last year when Plant Vogtle Unit 3 in Georgia entered commercial operation while Unit 4 started operations on April 29 of this year. However, we do not believe there are plans to build any additional nuclear plants in the US.

| EV ADOPTION

Load growth from EVs has been overlooked. Electricity consumption increases by 80 kilowatts (kW) with every EV that replaces an internal-combustion engine vehicle.⁷

The average total annual EV load in the US could rise from an estimated 24,000 gigawatts (GW) in 2023 to 468,000 GWh by 2040, according to PwC. That's an 1,850 percent increase or about as much as the total electricity generated annually in Texas. This EV load could be 9 percent to 12 percent of the projected US grid capacity, which is below current reserve

margins, but still could significantly impact load demand and generation needs.⁷

Given there are 280 million vehicles in the US and EV penetration is at 8 percent, electricity consumed by EVs is expected to grow to 80 percent by 2040 when EV penetration is expected to rise to 323 million which will demand 1000 GW to the grid, or 18 percent of total electricity demand.⁷

“As EV chargers are built out, not only do they draw a lot of power but the way they draw it is unique and terrifying. Two ultra-fast chargers that replicate the recharge time of an internal-combustion engine draw the same amount of power as a super Walmart,” said a former regulator.

“A grid operator knows when a Walmart is going to be drawing power and when it’s not. Plugging in different ultra-fast EV chargers at different places on

the grid at different times is the equivalent of having super Walgreens pop up randomly at random locations. If this demand is something that's going to come to pass, the amount of transmission interconnectedness, and substations which you would need to address that kind of volatility and establish the flexibility you need is mind-boggling."

TURBOCHARGED ELECTRICITY DEMAND

The clean energy transition, vehicle electrification, and the rise in GenAI have heightened demand for power and electricity over the next decade, pushing utilities, power generation providers, and grid operators to the limit.

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FOOTNOTES

PJM is a regional transmission organization that coordinates the movement of wholesale electricity in all or parts of the North Eastern interconnection grid operating in 13 states and the District of Columbia.

¹Gimon , Eric, O’Boyle, Mike, and Solomon, Michelle.

“Meeting Growing Electricity Demand Without Gas.” Energy Innovation. March 2024.

²Buddy Rizer, Executive Director for Economic Development in Loudoun County VA at the Morgan Stanley Powering the Growth of GenAI Symposium on March 26, 2024.

³Walton, Robert. “US data center electricity demand could double by 2030, driven by artificial intelligence: EPRI”, Utility Dive, May 30, 2024.

⁴Frequently Asked Question: What is U.S. electricity generation by energy source?” EIA Independent Statistics and Analysis, eia.gov, February 2024.

⁵“U.S. coal exports account for larger share of a shrinking market” EIA Independent Statistics and Analysis, eia.gov, January 29, 2024.

⁶“5 Fast Facts About Nuclear Energy”. Office of Nuclear Energy, energy.gov, June 11, 2024.

⁷McBride, Sam and Siokos, Konstantinos, “EV charging growth: How can power and utilities prepare?”, PWC.com

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